Scalar Field Dark Energy: ϕCDM Model

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Outline

- Introduction
- ΛCDM Model
- Alternative DE Model
- ϕCDM Model
- Data Analysis Techniques
- Analysis from Different Observation

Introduction

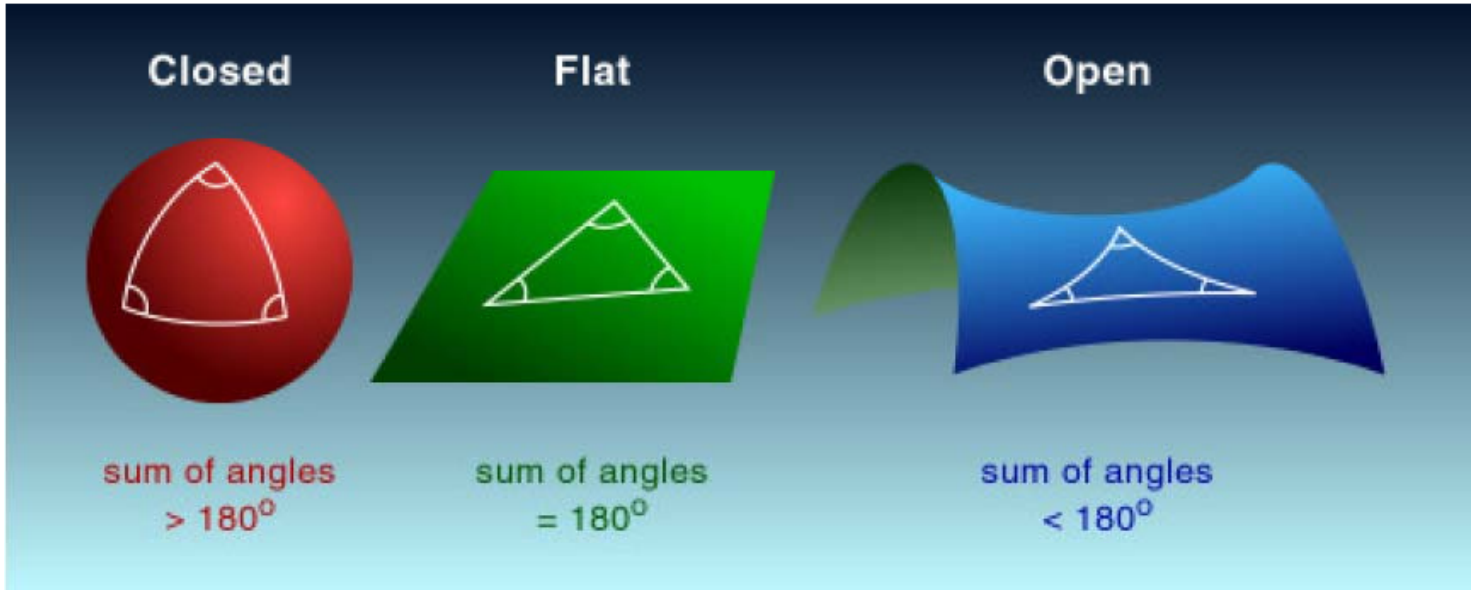
- Homogeneous and isotropic 4-dim space-time

$$\mathrm{d}s^2 = g_{\mu\nu}\mathrm{d}x^\mu \mathrm{d}x^\nu = -\mathrm{d}t^2 + a^2(t)\gamma_{ij}\mathrm{d}x^i \mathrm{d}x^j .$$

- In spherical coordinates

$$\gamma_{ij}\mathrm{d}x^i \mathrm{d}x^j = \mathrm{d}r^2 + \chi^2(r)\left(\mathrm{d}\theta^2 + \sin^2(\theta)\mathrm{d}\phi^2\right),$$

$$\text{where } \chi(r) = \frac{1}{\sqrt{-K}}\sinh\sqrt{-K}\,r.$$



Introduction (continued)

- Einstein equation

$$R_{\mu\nu} + \frac{1}{2} g_{\mu\nu} R = 8\pi G T_{\mu\nu}.$$

- Energy-momentum tensor for isotropic perfect fluid

$$T_{\mu\nu} = \begin{pmatrix} -\rho(t) & 0 & 0 & 0 \\ 0 & P(t) & 0 & 0 \\ 0 & 0 & P(t) & 0 \\ 0 & 0 & 0 & P(t) \end{pmatrix}.$$

- Energy conservation law

$$T^{\mu}{}_{\nu;\mu} = 0,$$

$$\Rightarrow \dot{\rho} = -3(\rho + P)H, \text{ where } H \equiv \frac{\dot{a}}{a} \text{ and } w \equiv \frac{P}{\rho},$$

$$\Rightarrow \dot{\rho} = -3(1+w)\rho\frac{\dot{a}}{a},$$

$$\Rightarrow \rho \propto (a_0/a)^{3(1+w)} \text{ and } a_0 = 1.$$

Introduction (continued)

- Equation of state
  - Radiation: $w = 1/3 \;\Rightarrow\; \rho_{\mathrm{r}} \propto \left(a_0/a\right)^4$.
  - Non-relativistic matter: $w \simeq 0 \;\Rightarrow\; \rho_{\mathrm{m}} \propto \left(a_0/a\right)^3$.

- From Einstein's equation
  - *t-t* component $\Rightarrow$ Friedmann equation: $\boxed{\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho - \frac{K}{a^2}}$

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho - \frac{K}{a^2} \;\Rightarrow\; H = H_0\sqrt{\frac{8\pi G}{3H_0^2}\rho - \frac{K}{a^2 H_0^2}},$$

$$\rho = \frac{3H_0^2}{8\pi G}\left[\Omega_{\mathrm{m}}\left(\frac{a_0}{a}\right)^3 + \Omega_{\mathrm{r}}\left(\frac{a_0}{a}\right)^4\right],\; \Omega_{\mathrm{K}} \equiv \frac{K}{a_0^2 H_0^2},$$

$$\rho_{\mathrm{m}}^{(0)} \equiv \frac{3H_0^2\Omega_{\mathrm{m}}}{8\pi G},\quad \text{and } \rho_{\mathrm{r}}^{(0)} \equiv \frac{3H_0^2\Omega_{\mathrm{r}}}{8\pi G},$$

$$\Rightarrow H = H_0\sqrt{\Omega_{\mathrm{m}}\left(\frac{a_0}{a}\right)^3 + \Omega_{\mathrm{r}}\left(\frac{a_0}{a}\right)^4 + \Omega_{\mathrm{K}}\left(\frac{a_0}{a}\right)^2}.$$

Introduction (continued)

- Acceleration is defined to mean that $\mathrm{d}^2 a / \mathrm{d}t^2 > 0$

- From *i*-*i* component Einstein equation

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}\left(\rho + 3P\right) > 0,$$

$$\rho + 3P < 0,$$

$$\Rightarrow w < -\frac{1}{3} \text{ (negative pressure } P < -\frac{\rho}{3}\text{)}.$$

Introduction (continued)

- The first evidence of accelerated expansion was provided by SNeIa absolute magnitude versus redshift data.
- SNe are classified based on the absorption lines of chemical elements in their spectra.
- SNe that do not have hydrogen atom Balmer lines in their spectra are classifed as Type I, while those that have Balmer lines belong to Type II.
- SNe can be farther divided into subtypes. SNe Type Ia (SNeIa) for example lack hydrogen lines but have a singly ionized silicon (Si II) line at 615 nm.

ΛCDM Model

- It is possible to add the cosmological constant term to the Einstein equation

$$R_{\mu\nu} + \frac{1}{2} g_{\mu\nu} R + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu}.$$

- The Friedmann equation

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho + \frac{\Lambda}{3} - \frac{K}{a^2} \Rightarrow H = H_0 \sqrt{\Omega_{\mathrm{m}}\left(\frac{a_0}{a}\right)^3 + \Omega_{\mathrm{r}}\left(\frac{a_0}{a}\right)^4 + \Omega_\Lambda + \Omega_{\mathrm{K}}\left(\frac{a_0}{a}\right)^2}.$$

- ΛCDM model

$$\text{choose that } \begin{cases} \Omega_\Lambda \simeq 0.7 \\ \left.\begin{array}{l} \Omega_{\mathrm{m}} \simeq 0.25 \\ \Omega_{\mathrm{b}} \simeq 0.05 \end{array}\right\} \Omega_{\mathrm{M}} \simeq 0.3 \\ \Omega_{\mathrm{r}} \simeq 0 \end{cases} \Rightarrow \Omega_\Lambda + \Omega_{\mathrm{M}} = 1 \text{ and } w_\Lambda = -1,$$

and corresponding to a flat spatial geometry,

$$\Rightarrow H = H_0 \sqrt{\Omega_{\mathrm{M}}\left(\frac{a_0}{a}\right)^3 + \Omega_\Lambda}.$$

ΛCDM Model (continued)

- Two big problems of ΛCDM (base on anthropic argument)
  - "Smallness" problem or "fine tuning " problem

$$\Lambda \approx H_0^2 = \left(2.1332h \times 10^{-42}\,\mathrm{GeV}\right)^2,$$

$$\Rightarrow \rho_\Lambda \approx \frac{\Lambda m_{\mathrm{pl}}^2}{8\pi} \approx 10^{-47}\,\mathrm{GeV}^4,$$

in QFT

$$\rho_{\mathrm{vac}} \approx \frac{k_{\mathrm{max}}^4}{16\pi^2}$$ and valid up to the Planck scale m_{pl},

$$\rho_{\mathrm{vac}} \simeq 10^{47}\,\mathrm{GeV}^4.$$

  - "Coincidence " problem: there is a short period of time when the energy density of matter and cosmological constant are comparible, i.e., the same order in ρ_{M} and ρ_Λ

Alternative DE Model

- Modified matter
  - Quintessence (slowly rolling scalar field)
  - K-essence
  - Perfect fluid models

- Modified gravity
  - $f(R)$ gravity (modified ΛCDM $\Rightarrow$ non-linear Lagragian density in terms of R)
  - Scalar-tensor theories (R couples to ϕ with the form: $F(\phi)R$)
  - Braneworld models

- Inhomogeneous universe model

ϕCDM Model

- Assume that a self-interaction scalar field ϕ minimally coupled to gravity ispresent on cosmological scale.

- The scalar field is described by the Lagragian density

$$\mathcal{L}_\phi = -\frac{1}{2} g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi - V(\phi).$$

- The equation of motion is

$$\partial_\mu \left(\frac{\delta \mathcal{L}}{\delta(\partial_\mu \phi)} \right) - \frac{\delta \mathcal{L}}{\delta \phi} = 0,$$

$$\Rightarrow -\Box\phi + V'(\phi) = 0.$$

- If the scalar field is close to spatially uniform

$$-\Box\phi + V'(\phi) = 0,$$

$$\Rightarrow \ddot{\phi} + 3H\dot{\phi} + V'(\phi) = 0.$$

ϕCDM Model *(continued)*

- The action of scalar field is

$$S=\int \mathrm{d}^4x\sqrt{-g}\left[\frac{1}{16\pi G}R+\mathcal{L}_\phi\right]+S_M,\ S_\phi=\int \mathrm{d}^4x\sqrt{-g}\mathcal{L}_\phi .$$

- The energy momentum tensor of the scalar field is given by

$$T_{\mu\nu}(x)=-\frac{2}{\sqrt{-g}}\frac{\delta\left(\sqrt{-g}\mathcal{L}_\phi\right)}{\delta g^{\mu\nu}}=\partial_\mu\phi\partial_\nu\phi-g_{\mu\nu}\left[\frac{1}{2}g^{\alpha\beta}\partial_\alpha\phi\partial_\beta\phi+V(\phi)\right],$$

$$\Rightarrow\begin{cases}\rho_\phi=-T_{00}^{(\phi)}=\frac{1}{2}\dot{\phi}^2+V(\phi)\\ P_\phi=T_{ii}^{(\phi)}=\frac{1}{2}\dot{\phi}^2-V(\phi)\end{cases},$$

$$\Rightarrow w_\phi=\frac{P_\phi}{\rho_\phi}=\frac{\dot{\phi}^2-2V(\phi)}{\dot{\phi}^2+2V(\phi)}.$$

ϕCDM Model *(continued)*

- The Hubble parameter should be rewritten as

$$\Rightarrow H = H_0\sqrt{\Omega_{\rm m}\left(\frac{a_0}{a}\right)^3 + \Omega_{\rm r}\left(\frac{a_0}{a}\right)^4 + \Omega_\phi + \Omega_{\rm K}\left(\frac{a_0}{a}\right)^2}$$

- We got Ω_ϕ is the time-dependent energy density parameter of the scalar field
- A specific potential $V(\phi)$:
  - Self-interaction and inverse-power law potential:

$$V(\phi) = \frac{\kappa}{2G}\phi^{-\alpha} \propto \phi^{-\alpha}, \text{ where } \alpha>0,$$

$$\rho_\phi = \frac{1}{2}\left(\dot{\phi}^2 + \frac{\kappa}{G\phi^\alpha}\right),$$

$$\begin{cases} \left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\left(\rho_{\rm m} + \rho_\phi\right) \Rightarrow \frac{\dot{a}}{a} = H_0\sqrt{\Omega_{\rm m}\left(\frac{a_0}{a}\right)^3 + \Omega_\phi} \\ \ddot{\phi} + 3\frac{\dot{a}}{a}\dot{\phi} - \frac{\kappa}{2G}\frac{\alpha}{\phi^{\alpha+1}} = 0 \end{cases}.$$

ϕCDM Model *(continued)*

- Larger values of α correspond to faster evolution of the scalar field.
- $\alpha = 0$ corresponds to the time-independent cosmological constant.

$$V(\phi) = \frac{\kappa}{2G}\phi^{-\alpha}.$$

- Tracker solution ($\rho_\phi \ll \rho_{\mathrm{m}}$), in the distant past:
  - Assume that the universe is spatially-flat, accepted solution is

$$\phi \propto a^{3(1+w_m)/(\alpha+2)} \Rightarrow w_\phi = \frac{\alpha w_m - 2}{\alpha + 2}.$$

- The coupled differential equations must be solved numerically

$$\begin{cases} \dfrac{\dot{a}}{a} = H_0\sqrt{\Omega_{\mathrm{m}}\left(\dfrac{a_0}{a}\right)^3 + \Omega_\phi} \\ \ddot{\phi} + 3\dfrac{\dot{a}}{a}\dot{\phi} - \dfrac{\kappa}{2G}\dfrac{\alpha}{\phi^{\alpha+1}} = 0 \end{cases}.$$

ϕCDM Model *(continued)*

- The numerical algorithm:
  - 4th order Runge-Kutta method

- For $\alpha = 0$ the ϕCDM model reduces to the spatially-flat ΛCDM with $\Omega_\phi = \Omega_\Lambda$